

Deva Anand

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Education

University of Massachusetts Amherst

Master of Science in Computer Science

September 2025 – May 2027 (Expected)

Amherst, MA

Coursework: Systems for Data Science, Performance Engineering, Optimization in CS, Advanced Machine Learning, Advanced NLP

Vels Institute of Science, Technology and Advanced Studies

Bachelor of Engineering in Computer Science and Engineering

August 2019 – May 2023

Chennai, India

Skills

Languages

Python (typing, async), JavaScript, TypeScript, SQL, Java, C++, Bash, HTML/CSS, Tailwind

AI-Assisted Dev & Agents

Claude / Claude Code, GitHub Copilot, Cursor, ChatGPT / Codex, agent-driven workflows, LLM agents, OpenAI / Groq APIs, RAG

Backend & APIs

FastAPI, Node.js, Express, REST APIs, async services, query optimization, caching, integration testing

Cloud & DevOps

AWS (S3, KMS), GCP, Azure, Docker, Git, GitHub Actions, CI/CD, git-webhook auto-deploy, Linux

Databases & Data

PostgreSQL, MongoDB, Redis, SQLite, Apache Spark / PySpark, Spark UI, React

Experience

Wysa

Backend Engineer

July 2023 – July 2025

Bengaluru, India

- Engineered **multi-tenant NHS eTriage platform** for **8+ UK clients** (Node.js, MongoDB) processing **10,000+ monthly submissions** as a production-grade system integrated with Mayden's iaptus platform via REST APIs and aggregation pipelines, contributing to **\$340K+** in revenue.
- Architected full-stack data annotation platform (**React, Node.js, MongoDB**) used daily by AI team to label data for production models, with assets/files stored in **AWS S3** (multi-region buckets); cut manual operations by **100+ hours/month** and improved labeling throughput by **60%**.
- Joined a **GitHub Copilot** productivity pilot, routing tech-debt PRs, code migrations, and reviews through AI-assisted workflows and maintaining a versioned conventions file to keep agent-generated code aligned with team patterns, while owning backend reliability and debugging production incidents across releases.

Cario Growth Services Pvt. Ltd.

Backend Developer Intern

November 2022 – May 2023

Chennai, India

- Designed and implemented **20+ RESTful APIs** in Node.js and Fastify, backed by **PostgreSQL**, with input validation, clear contracts, and backward-compatible changes used in production by the frontend team.
- Migrated core codebase from **JavaScript to TypeScript** across multiple repositories and prototyped LLM-powered context-aware replies in a chat product, improving type safety, maintainability, and end-user response quality.

Projects

AutoEval - Model Evaluation Platform

github.com/Deva-1903/AutoEval

Python, ML Evaluation, Git Automation

- Built an **evaluation-improvement agent** that proposes test cases and labels, scores them against multiple systems, and accepts only updates that improve **discriminative power** on a holdout-heavy fitness objective, exactly the kind of evaluation-platform tooling a research team relies on.
- Added anti-Goodharting guardrails (schema checks, duplicate rejection, read-only holdout protection) and committed accepted updates back to Git with report cards for an auditable, reproducible evaluation workflow.

AgenticSearch - Provenance-First AI Pipeline

github.com/Deva-1903/ciir_agentic_search

Python, FastAPI, OpenAI / Groq, SQLite

- Developed a multi-stage **agent-driven** service (typed retrieval planning, extraction with deterministic parsers before LLM fallback, output verification, cell-level provenance), built end-to-end with AI-assisted development.
- Hardened the FastAPI service with **150+ automated tests**, provider routing/fallback, and pipeline instrumentation for reliable, debuggable production behavior.

Spark ETL Performance Optimization (NYC TLC Trip Data)

github.com/Deva-1903/Spark-ETL-Optimization

Apache Spark / PySpark, SQL, Python

- Cut end-to-end ETL runtime roughly **25%** (17 to 19 min down to 14.3 min) on a **1.4B-row** dataset via **query optimization**, broadcast joins, adaptive execution, and partition tuning, and reduced analytics tail-latency ratio (P99/P50) from **2.14x to 1.74x** via partition/column pruning and filter pushdown.

Publication & Extracurriculars

- Publication:** "Alzheimer's Disease Classification using Transfer Learning," **IEEE CONIT 2023**, applied deep transfer learning on neuroimaging data for medical image classification.
- Selected for **GirlScript Summer of Code (GSSoC) 2023** and contributed to multiple open-source projects (Linkfree, Freehit, ProjectsHut).
- Tutored **30+** underprivileged students in programming at Sayur, non-profit in South Tamil Nadu.